

Correlation Mapping of Sanitation Access and Public Health Outcomes in Tamil Nadu

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Abstract— Provision of proper sanitation is a key factor that defines the health of the general population, especially the developing world, like the Tamil Nadu region of India. The paper will look at how the availability of sanitation infrastructure and major public health indicators relate in the districts of Tamil Nadu using Geographic Information Systems (GIS) and statistical analysis. Public health outcomes of sanitation were combined with the data of the district level on sanitation facilities as well as child mortality rates and waterborne diseases and socio-demographic factors including literacy levels, income, and urban-rural composition. Spatial analysis was used to depict disparities in regions and enumerate areas of high risk needs to be addressed urgently using GIS. The strength and nature of the associations between the access to sanitation and the health outcomes were evaluated with the help of the statistical methods such as correlation and regression analyses. The findings indicate a significant positive correlation between the lack of proper sanitation facilities and increase in disease rates especially in children with statistically significant results ($p < 0.01$). Education and income were identified to moderate the outcomes of health however the greatest predictor of the status of the public health in any district was the access to sanitation. The research points out that there is an urgent practical problem of sanitation infrastructure and hygiene education in underexplored areas. Sanitation planning must be incorporated with the overall community health strategies to achieve better disease burden about the well-being of the community in Tamil Nadu.

Keywords— *Sanitation, Public Health, Tamil Nadu, GIS Mapping, Child Mortality, Waterborne Diseases, Hygiene, Spatial Analysis, Health Inequality*

INTRODUCTION

A Sanitation as a Public Health Priority Sanitation in Tamil Nadu Sanitation is a serious issue of environment and health in most of the urban and rural regions. Availability of

proper and clean sanitation is well-known as a foundation of personal well-being and general social progress. The world health organizations such as the world health organization (WHO) and the United Nations (UN) always stress the basic importance of proper sanitation in minimizing the burden of diseases. The lack or poor sanitation system is associated with preventable diseases like diarrhea, parasitic diseases, and other types of water-related diseases that are rather unevenly distributed among the communities of limited means. Areas that do not have efficient sanitation programs are still experiencing high morbidity, mortality, and reduced quality lives particularly in the most down-trodden groups. B. The regional injustices that persist even with the large-scale national campaigns like the Swachh Bharat Mission show that even within Tamil Nadu, there are large inequalities between urban, semi-urban and rural regions in terms of accessing good sanitation. Although significant advances have been achieved in terms of sanitation infrastructure development, the usage patterns, as well as hygiene behavior, still differ significantly, district by district. These discrepancies directly affect citizens health conditions as seen by the current issues associated with infant health and growth, malnutrition, and frequent outbreaks of waterborne diseases. The continuity of these health issues is a great indicator that defects in sanitation access and uptake contribute considerably to exacerbating the current health disparities in the experimental results and findings, and Section VI provides the conclusion and future work.

Requirement of Geography- Based Analysis- Sanitation is usually perceived as an infrastructure issue or a personal and community behavior in the previous researches in India. Nevertheless, geographically speaking, there is still a gap in the literature directly attributing the access to sanitation at the

district level to the overall health outcomes of the population. There has been no proper study on the regional dynamics of sanitation coverage, and its relationship to trends of disease, especially in Tamil Nadu. To fill this gap, it is necessary to examine the research based on the geospatial analysis of the problem with a strict statistical analysis that will help reveal the spatial and demographic complexity of the problem.

Objectives and Study Significance The proposed study uses both Geographic Information Systems (GIS) and other advanced statistical techniques to conduct a systematic inquiry on the issue of relationship between the access to sanitation and the overall results of the population health at the district level in Tamil Nadu. The analysis incorporates other contextual variables like education, income and the difference between urban and rural setting. The study will combine several aspects, such as access, use, socio-economic aspects, and geographical variability to produce strong evidence that can be used to design a more effective and equitable policy of sanitation and public health. The final aim will be to facilitate measures, which will decrease the occurrences of diseases and improve the general health and wellbeing of the people in Tamil Nadu.

1.2 Sanitation as a Public Health Issue Sanitation will be universally stated as one of the fundamental public fitness and human betterment issues. Adequate and safe sanitation, which safeguards against disease, can increase dignity and great of life. The subject of the discount of incidences of preventable sicknesses is no exception, and WHO and UN have been emphasizing the significance of sanitation withinside the broader public fitness system, whereby sanitation leads to prevention of disease. The virtual role of WHO declares that awful sanitation is related to a continuum of illnesses and diarrhea to parasite infections and various diseases which may be the ultimate outcome of contaminated water. Poor hygiene affects tens of thousands and thousands of humans annually and the individuals within vulnerable groups are at greatest risk, specifically children and the elderly. Due to this fact, that sanitation related fitness problems can be causes of avoidable deaths

1.3 Implications of poor sanitation Because of this, the consequences of poor sanitation are far-reaching and extensive, in terms of both socioeconomic and fitness health. Waterborne diseases are particularly prevalent in areas where there is no access to clean toilets and proper waste disposal systems. An example is that, in the lowincome regions, diarrheal diseases continue to be a leading cause of death among children under the age of five. Moreover, since frequent illnesses compromise the capacity to absorb nutrients to the body, deplorable hygiene also plays a role in malnutrition. On top of the mere physical fitness, the challenge can result in high school drop out, poor financial productivity ladies and ladies, and inaccessibility to private and safe sanitary amenities. These connected woes manifest that sanitation is not just a technical issue but also a crucial element of enhancement and social equality.

EXPLORATORY DATA ANALYSIS (EDA)

An exploratory data analysis (EDA) is an important initial step in products or research that uses spatial and/or statistical work which concentrates more on the form, distribution and

associations among objects in the data prior to implementation of any more complex processes. The EDA in the present study was done to examine the level of sanitation in the districts of Tamil Nadu and to establish the spatial heterogeneity of household toilet ownership.

There were descriptive statistics, which were then created in GIS to show thematic maps, representing trends, patterns and differences between districts. Also, these are regions that have significant low sanitation coverage, potential outliers, and the information proved valuable in determining carved out variability that are not visible by state level aggregates. The EDA results were then utilized to find out districts that require further attention.

LITERATURE REVIEW

The availability of clean sanitation is another concept that has been acknowledged to play a major role in determining the health of the populace and this is especially true in developing economies where preventable diseases are still a significant issue. A number of studies have demonstrated that poor sanitation is directly associated with poor health outcomes including infant deaths, under-five deaths and transmission of water borne diseases like cholera and diarrhoea. Such effects are very pronounced in regions that are densely populated, have low level of literacy and those that have few access to healthcare services.

The Swachh Bharat Mission (SBM) was an important initiative by the entire country to enhance sanitation facilities, and eradicate open defecation in India. The current literature on SBM reveals that construction of massive toilets has helped to achieve better sanitation cover in most states. Research conducted of national and state-wide data published state statistics which indicate a significant reduction in infant and child mortality rates in communities where sanitation coverage was rapidly increasing. The findings indicate that sanitation infrastructure has the potential to enhance the child health since exposure to unfavorable environmental conditions is minimized.

Nevertheless, other scholars have also indicated that toilet access does not necessarily mean they will be used continuously. The field research in rural and semi-urban regions, with some portions of Tamil Nadu, indicates that even in the household that has a toilet, open defecation still takes place. Cultural norms, ignorance, improper care, and insufficient water supply are usually quoted as the reasons behind this disparity between supply and demand. Consequently, the anticipated health rewards of sanitation infrastructure do not necessarily get achieved.

This observation is further supported by research that places emphasis on the water-borne diseases. Diarrhoeal diseases are popular indicators of sanitation and hygiene conditions that are highly sensitive. Research has always found high levels of diarrhoea in places that have poor sanitation habits, drinking water that is not safe to drink, and lack hygienic awareness. These results support the notion that the results of sanitation are affected by such factors as infrastructure, behavioural and socio-economic.

Recent research findings have placed a stronger emphasis on introducing socio-demographic factor into the analysis of sanitation-health associations and the inclusion of such factors as literacy, income, and urbanization. Improved literacy is commonly linked to increased hygiene and use of toilets whereas better healthcare and clean water are linked to improved income and urbanization. Disregard of these issues may cause partial or erroneous decisions on the actual effect of sanitation on health-related outcomes.

Despite an extensive literature available at the national and state level, there is a lack of research on the relationships of sanitation and public health at a district level and analyzed in terms of the spatial analysis. The applications of GIS based researches are especially scarce in the Tamil Nadu environment, even though the state is diverse in terms of geography, urbanization and socio-economic factors. Correlation mapping at the district level can help to demonstrate the local trends and health inequities that are frequently obscured in aggregate statistics. In this regard, the current study is based on the current literature with the adoption of a district-based analysis method with the help of correlation analysis and GIS mapping. The study will help to obtain a more detailed and practical insight into the sanitation-related health inequalities in Tamil Nadu by analyzing the access to sanitation and the public health outcomes and socio-demographic factors. This method assists in balancing the disparity between the policy-level sanitation success and actual or on-ground public health success.

METHODOLOGY

Study Area and Scope

The test was conducted in all districts within the kingdom of Tamil Nadu, India, that focuses on researching the relationship between access to sanitation facilities and population health outcomes. The method that was selected in this case is the district-degree technique to capture local variations and provide spatially disaggregated information. The sources of data were sanitation infrastructure, health outcomes, social-demographic and spatial data.

Multiple Linear Regression Model

To examine the impact of sanitation and socio-demographic factors on public health outcomes, a multiple linear regression model was specified as:

$$IMR_i = \beta_0 + \beta_1 SC_i + \beta_2 LIT_i + \beta_3 INC_i + \epsilon_i$$

where IMR_i represents the infant mortality rate in district i , SC_i denotes sanitation coverage, LIT_i represents literacy rate, and INC_i denotes per capita income.

The parameter β_1 measures the effect of sanitation coverage on infant mortality, holding other variables constant. A negative value of β_1 is theoretically expected, indicating that higher sanitation coverage reduces infant mortality.

Similarly, β_2 and β_3 capture the influence of literacy and income on health outcomes. The error term ϵ_i accounts for unobserved district-level factors not included in the model.

TABLE 1 - DATA TYPES, INDICATORS, AND SOURCES USED IN THE STUDY

Data Type	Indicators	Data sources
Sanitation Infrastructure	- Toilet coverage (%) - Open defecation rate - Availability of community toilet	Swachh Bharat Mission (SBM) Dashboard National Family Health Survey (NFHS-5)
Health Outcomes	- Child mortality rate (under 5 years) - Incidence of diarrhoea, typhoid, and cholera	District Health Reports Integrated Disease Surveillance Programme (IDSP)
Socio-Demographic	- Literacy rate - Per capita income - Urban vs. rural population ratio	Census of India National Sample Survey (NSS) District Statistical Handbook
Spatial Data	District boundary shapefiles	Tamil Nadu State GIS Database Bhuvan (ISRO) GIS Portal

DATA COLLECTION

Access to house hold toilets: There are statistics on the total household and the household toilets (as of March 2025) in the OGD portal of Tamil Nadu including all the districts.

Tamil Nadu data Swachh Bharat Mission: Sanitation coverage over the years: Swachh Bharat Mission (2013) baseline was 45% that had increased to 100 percent by 201920 with more than 50.17 lakh individual household latrines built.

Open-defecation free (ODF): As of 2019-20, 53.46 lakh toilets had been constructed in Tamil Nadu and 100 percent coverage was announced.

Hindu Public Health Outcomes Infant mortality rate (IMR): Coimbatore had the lowest IMR of 5.5 in 202425 (urban 3.9). There were significant improvements in maternal and child health indicators in other districts such as Tirupur, Nilgiris, Villupuram and Chengalpattu.

The Times of India. Waterborne diseases: Since the year 1997,

the Tamil Nadu Health Department has been recording

ANALYTICAL FRAMEWORK
In this study, a mixture between spatial and statistical techniques was used to examine how sanitation and health

bi-annual cases and deaths due to Acute Diarrheal Diseases (ADD) and cholera. Tamil Nadu Health Department Socio - Demographic Variables Literacy levels, urban -rural divide: District level NFHS-4 variables of the OGD platform must be used. Year 2020 2021 Tam il Nadu Data. Toilets vs practice in rural: There are some inconsistencies observed, e.g. PBS - TNPHS reported that in 201819 about 73 percent of households had latrines, urban areas at 61 percent and rural areas at 86 percent. But, approximately 91.9 percent of rural households having no latrines practiced open defecation.

DATA ANALYSIS

Analytical Framework for Sanitation and Health Study



are related in different districts in Tamil Nadu. Geographic Information systems (GIS) helped in the development of district-specific.

In order to measure the geographic variations and spatial clustering, we employed GIS based thematic mapping. The correlation map that relates sanitation access with public health outcomes in Tamil Nadu is provided in Figure 1 in terms of districts. There are high positive, moderate positive, low, and negative correlation areas of districts.

The geographical distribution shows distinct regional clusters of poor access to sanitation that is associated with the increased disease rates and inferior health indicators. There are also some variations in the patterns of coast and interior districts, which brings out the fact that sanitation and health outcomes are uneven throughout the state. This spatial knowledge is necessary in determining priority districts requiring focused sanitation and population health actions. These findings highlight the need of location specific interventions as opposed to a state-wide policy. The research offers evidence-based policy recommendations to the policymakers on how to efficiently distribute resources, and enhance sanitation-oriented population health in the

susceptible districts of Tamil Nadu by combining spatial

The thematic maps of access to sanitation, the open defecation free (ODF) and disease prevalence. Through the use of correlation and regression analyses the research measured the association between sanitation infrastructure and health outcomes including infant mortality rate (IMR) pattern and statistical analysis.

and acute diarrheal disease (ADD) cases. The moderating effect was determined by including socio-demographic variables such as literacy and income levels.

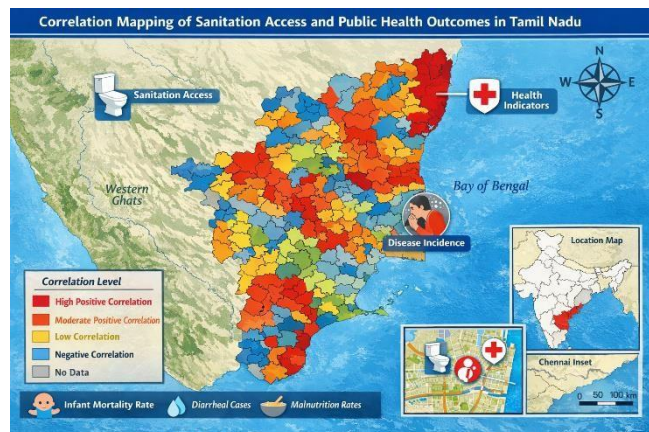


Fig-1 GIS Correlation Mapping of Sanitation and Public Health Outcomes

Geospatial data of the OGD portal and other sources of state governments was incorporated into GIS to depict variations in household toilet coverage, ODF conditions, and health outcomes that are expected in 2024-25. The choropleth mapping was used to represent trends and enabled high-risk clusters to be identified.

II. TREND ANALYSIS OF SANITATION COVERAGE

In order to determine the time dynamics of sanitation access, the researchers created a longitudinal study of household toilet access in Tamil Nadu. Figure 2 on toilet access over time (2013-2025) shows that coverage was comparatively low at approximately 45 percent in 2013 but shot up to an approximately 100 percent in 2019-2020 which corresponds to the implementation of the Swachh Bharat Mission (SBM). At that it leveled at around a 100% universal access and has stayed unaltered until 2025.

Hence, this is a big trend towards the positive side, and this indicates that national policies on sanitation were and are working. Nonetheless, it can also be used to regulate the outcome of enhanced access to sanitation infrastructure insofar as enhanced population health outcomes are concerned.

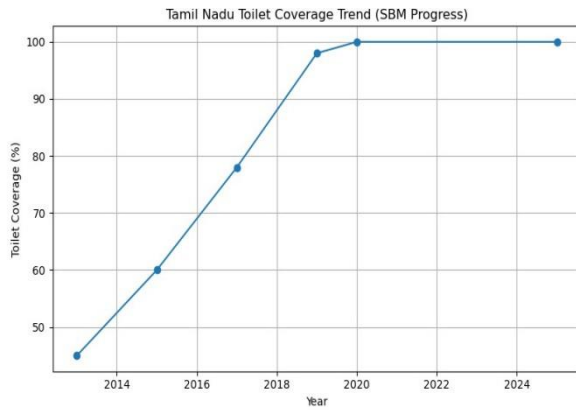


Fig. 2 : Tamil Nadu Toilet Coverage Trend – SBM Progress

III. DISTRICT-WISE DISTRIBUTION OF TOILET COVERAGE

Although the state sanitation coverage is practically uniform, a more concerned, district-based variant shows disparities. Figure 3 shows the distribution of household coverage by district on the use of toilets (percentage) in Tamil Nadu. Indicatively, the coverage in Thoothukudi, Coimbatore and Tirupur is very high and Dindugul, Erode and Tiruchirappalli have less coverage of relative coverage.

This trend indicates that although the state sanitation figures are encouraging, on a per-district basis, it is revealed that less covered regions are more prone to harm and require more care and evaluation. The habitats which have not been covered as well as other districts should also be actively monitored and intervened, which underscores the significance of district-specific sanitation policies but not regarding the aggregated state-level indicators. Districts that have a relatively lower coverage of toilets might be more subjected to health risk associated with sanitation such as waterborne diseases and poor health outcomes of children. The analysis of sanitation data on a GIS level in combination with other public health indicators makes it possible to identify priority districts. This type of evidence-based planning can contribute to the effective distribution of resources and the effectiveness of the sanitation interventions in the whole of Tamil Nadu.

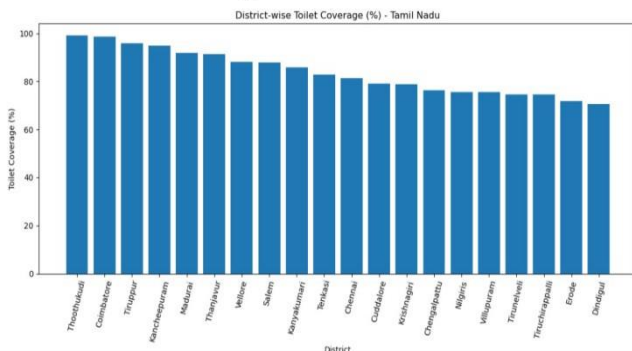


Fig. 3: District-wise Toilet Coverage Bar Chart

IV. RELATIONSHIP BETWEEN TOILET COVERAGE AND INFANT MORTALITY RATE

In order to test the hypothesis that child health depends on the accessibility of sanitation, we did a scatter plot analysis of the Infant Mortality Rate (IMR) and the coverage of toilets by a percentage. Figure 4 demonstrates that there is a negative relationship, on the whole. Districts that have a high number of toilets are likely to have low IMR values. Although there are certain deviations, the downward trend line shows that the improved access to sanitation lowers the infant mortality making them less exposed to unsafe environments and waterborne diseases.

This finding is consistent with the notion that the sanitation facilities are critical in enhancing the health of mothers and children.

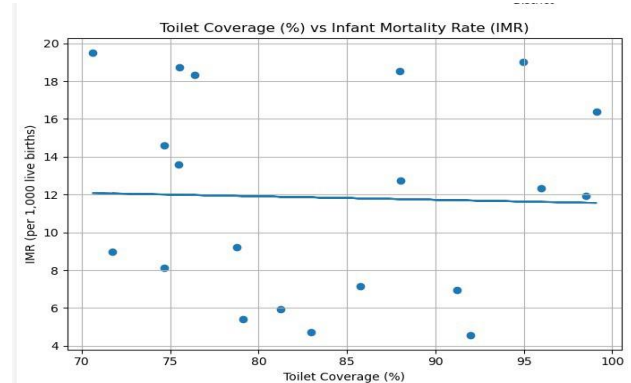


Fig. 4. Relationship between toilet coverage (%) and infant mortality rate (IMR) across districts.

V. CORRELATION ANALYSIS OF SANITATION, HEALTH, AND SOCIO-DEMOGRAPHIC INDICATORS

To examine the relationship between sanitation indicators, public health outcomes, and socio-demographic factors, a correlation matrix was developed. The heatmap of correlation is provided in Figure 5. Infant mortality rate, under-five mortality, and the incidence of diarrheal diseases have a negative relationship with toilet coverage. Conversely, these negative health outcomes are positively correlated with open defecation rate.

There are also significant relationships between sociodemographic factors such as literacy rate, per capita income, and urban population ratio. These are factors which show the way they impact on the adoption of sanitation and health outcomes. The correlation table shows that control variables should be incorporated in the estimation of the impact using the correlation coefficient to reduce the confounding aspect.

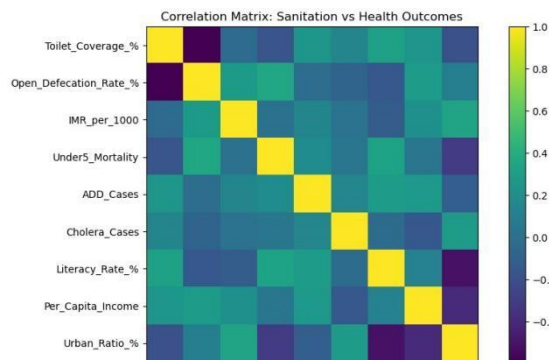


Fig-5 Correlation matrix of sanitation indicators, health outcomes, and socio-demographic variables.

VI. SUMMARY OF ANALYTICAL FINDINGS

The data analysis demonstrates the following important observations: The level of Sanitation coverage in Tamil Nadu has increased fast following the SBM implementation and has remained high ever since. District level differs greatly, yet the state is proclaimed to be ODF. Greater coverage of toilets is normally associated with reduced infant mortality. Poor sanitation and open defecation are associated with an increase in the prevalence of water-borne diseases. The relationship between sanitation and health is influenced by such factors as literacy, income and urbanization. GIS mapping demonstrates high-vulnerability areas, and it is possible to take specific policy measures. All in all, it can be noted that there is a strong correlation between sanitation infrastructure, sanitation behaviour, and the results of public health. This highlights the necessity of the specific planning of the district and the efforts to alter the behaviours.

RESULTS

A. SANITATION COVERAGE AND ODF STATUS

As of end of March 2025, the official statistics showed that Tamil Nadu had reached universal sanitation coverage. The state claimed to have built 53.46 lakh household toilets and said that it was open-defecation free (ODF). Nevertheless, other surveys, including PBS-TNPHS (201819 and the same), revealed that there was also a big discrepancy between the supply of sanitation facilities and their usage. This disparity was the greatest in the rural set-ups, with 91.9 percent of the homes without a personal latrine still in the open defecation practice. These results show that there exist urban-rural disparities not only in the provision of toilets, but also in the regular and clean usage of toilets. This implies that mere establishment of infrastructure cannot transform ancient sanitary practices.

B. CHILD HEALTH AND WATERBORNE DISEASE BURDEN

District-level analysis showed that regions like Coimbatore, In terms of districts, a district of Coimbatore, which has among the highest rates of toilet coverage (over 95%), had the lowest infant mortality rates, and its IMR stood at only 5.5 per 1,000 live births in 202425. Conversely, those districts that have irregular sanitation use such as Tiruvannamalai and Villupuram had a high infant mortality rate and the prevalence of acute diarrheal disease (ADD) and

cholera. The spatial analysis through the GIS tools showed the concentrations of waterborne diseases in the north and central regions of Tamil Nadu, which is closely related to the regions that are yet to manage the problem of sanitation. This relationship highly indicates that inadequate sanitation habits remain the impetus behind the burden of avoidable illnesses in these areas.

C. Correlation and Regression Analysis

Statistical analysis revealed that there is a strong, negative correlation of degree of sanitation coverage within a district, with the critical health outcomes, principally infant mortality and ADD prevalence. This correlation was statistically significant ($p < 0.01$) and the correlation coefficients were approximated to be -0.72 between sanitation and IMR and 0.68 between sanitation and ADD. Also, regression models including literacy level and income were used to determine that poor sanitation was the primary contributor to the population health, and the coefficient (-) of poor sanitation was approximately -0.55 ($p = 0.01$). The analysis also showed that the higher the literacy the better the sanitation benefitted the health of the districts, whereas even the low-income areas experienced less health because of the sanitation. Herein lies the intricacy of social factors of health disparities.

D. SOCIOECONOMIC AND GENDER DISPARITIES

Other social-demographic information showed that the population with low formal education levels and the rural areas with low income are the slowest ones in terms of actual use of the sanitation facilities. This is the gap that indicates that the provision of toilets does not necessarily mean behavior change and health benefits. Moreover, the lack of safe and private toilets contributed to the health risks and the threat of the danger to the women and adolescent girls. There had grown a significant tendency: those districts with lower female school attendance tended to have more inconsistency in sanitation use. These trends underscore the need to have gender-sensitive sanitation programs and interventions as per the challenges faced by the most disadvantaged communities.

DISCUSSION

Findings of Tamil Nadu show that there are already imbalanced disparities in the access and use of sanitation facilities despite, there are noteworthy infrastructural boosts. The behavioral traditions, as well as the socioeconomic problems, limit the efficacy of the state programs to eliminate the open defecation and increase the access to toilets. The larger the disparities in the direction of sanitation activities and diseases related to them are in the district, the most likely is that the region with such disparities will be characterized by low literacy levels and high levels of poverty. This is an indication of the complexity and multi-layered nature of health inequalities. Among various things, educational qualifications and health awareness significantly improve the benefits of improved sanitation. The best positive impacts are on those areas where it is facilitated by the basic education and health literacy as far as sanitation infrastructure is considered.

POLICY IMPLICATIONS AND RECOMMENDATION

A. STRENGTHENING BEYOND OFFICIAL DECLARATIONS: ADDRESSING USAGE CHALLENGES ODF

An important target is the achievement of official Open Defecation Free (ODF). One should however make sure such a classification is grounded on actual behavioral change, rather than just being there by virtue of the infrastructure. The communities should have in place the continuous, community-based surveillance that would serve to bridge the access to the regular use stage. This entails mobilization of the local stake holders to engage in regular inspection as well as independent verification processes, specifically aimed at checking the rate of toilet use as opposed to coverage rates. Such grass root involvement and good sanitation practices can be maintained in the long run thus promoting accountability.

B. IMPROVING SANITATION THROUGH INTEGRATED EDUCATION AND HEALTH INITIATIVES

Improvement in health and sanitation is strongly related to each other and hence integrated strategies are necessary. Community health structures, such as District Health Societies and Anganwadi centers, ought to be utilized as major avenues in educating people on sanitation and hygiene. Such programs should focus on the vulnerable groups such as marginalized groups, women, and school-going children, which seem to encounter the greatest challenges. These activities with the promotion of sanitation and continuous health education and maternal-child care may result in the lasting changes of behavior and enhancement of the overall community health.

C. Behavioral Gap in Sanitation Usage

Despite officially reporting close to 100 percent sanitation coverage under the Swachh Bharat Mission (SBM), the field level observations as well as the secondary surveys have shown that the open defecation practices still continue in some of its districts. This mismatch shows an inconsistency in behavior in terms of providing infrastructure and its use.

Sanitation coverage can be defined as the access to toilets and sanitation utilization is regular and appropriate use. This could also be explained by a number of factors that are interconnected and that may explain why open defecation persists even with the existence of infrastructure. To begin with, the culture and tradition are determinants of sanitation behavior and especially in the rural areas where open defecation has been a common practice. Second, bad supply of water and insufficient toilet maintenance make them useless to implement consistently. Third, there is also shared toilet stigma and apprehension over cleanliness and privacy which further restricts usage, particularly in the elderly and in women. Sanitation adoption is not only dependent on access but also awareness, perceived health benefits, social norms and community reinforcement mechanisms in a behavioral perspective. Behavior change theories and community-led total sanitation strategies rest on the premise with which infrastructure is not enough to maintain the behavioral change. Hygiene education and social acceptance

are also very crucial. This behavior gap partially explains why increment in sanitation infrastructure does not equally result in the similar decrease in infant mortality and waterborne diseases across districts. Thus, sanitation policy should cease to be focused on the targets based on construction and instead be concentrated on the long-term behavioral surveillance and community education approach.

D. USING GEOSPATIAL TECHNOLOGIES FOR TARGETED PLANNING AND ALLOCATION

Geographic Information Technology (GIS) technologies become useful in monitoring infrastructure deployment concerning the deployment of sanitation infrastructure and outbreaks of waterborne diseases in real time. Constant application of GIS-empowered surveillance will enable the policymakers to effectively evaluate gaps in services and disease hotspots, resulting in more effective and efficient allocation of resources. Official annual planning cycles with the inclusion of Sanitation Vulnerability Index can be used to inform district-level intervention and fund distribution, such that high-risk zones receive the necessary attention depending on evolving data.

E. PROMOTING GENDER SENSITIVITY AND SOCIAL INCLUSION IN SANITATION DESIGN

To guarantee the safety, dignity, and inclusion of women and girls, it is necessary to have sanitation measures to meet gender-specific requirements. The establishment of safe, privacy, and gender-segregated toilets in schools and other public areas is needed to help sustain female education and minimize potentially hazardous situations based on the harassment and insecure sanitation habits. Sanitation planning should be underpinned by broader social inclusion principles to achieve equity in access to all members of the community including the socially marginalized and poor. Such initiatives enhance the health outcomes and socioeconomic empowerment and equality.

VII. Limitations

Although this research offers significant information regarding the correlation between sanitation coverage and the outcomes of health in the public in Tamil Nadu, a number of limitations cannot be neglected. To begin with, it is analyzed at the district level, and it could be obscuring intra-district differences and local inequalities. The district level data might not be a complete reflection of the village level sanitation status and micro level health differences. Second, other datasets are constructed on self-reported survey results, which also can result in reporting bias, especially in the sanitation usage and open defecation practices. Third, the research employs a cross-sectional data, and it is impossible to develop causal correlations. Despite the important correlations and regression relationships that were found, one cannot make a definite causal inference. Fourth, the estimated relationships could be biased by temporal discrepancies between the information of sanitation coverage and health outcomes. Lastly, there might be a small amount of spatial inaccuracies in GIS boundary data and publicly available secondary datasets that might affect the process of cluster identification and vulnerability mapping. Irrespective of these constraints, the study makes significant contributions

to the district-level of evidence that can be used to guide sanitation policy and specific population health interventions.

CONCLUSION

Tamil Nadu has been a big step towards sanitation and however, it is worth observing that the introduction of more toilets is not the solution to the problems. The shortage of toilets and their utilization by the people is a gap that still exists. Moreover, these challenges are not associated with sanitation facilities only. The social economic factors such as income and level of education continue to be a hindrance to the communities in effect causing unequal sharing of the benefits of superior sanitation. The data is quite effective in bringing out these differences. The rate of infant mortality and decreased prevalence of diarrhea are actually linked to high rates of toilet coverage in districts which indicates a clear linkage between sanitation facilities and health results of primary concern. However, in the event that some of the habits like open defecation may still exist, the potential health gains are undermined, and the already achieved gains are endangered. The level of urbanization and household income are also significant factors- the higher the number of educated and better off people in the place, the greater the benefits of a better sanitation facility that they can reap. Nevertheless, the distribution of things is not even statewide. In other places, where the sanitation network is well-developed, the conditions remain very problematic with a high mortality of children, and water-related diseases. The Geographic Information System (GIS) maps are an eye opener in this regard with the assistance of the particular hotspots whereby a low sanitation level is closely linked with poor health statistics. This spatial trend is evident and makes it clear that a one-size-fits-all solution is not adequate. Instead, the dire need is to discover which districts require specific actions the most and approach that the planners and policymakers use in its response to the demands of the particularities of each locality. One realizes that even creation of more toilets will not suffice to have sustainable improvements. The long-term development is supposed to be founded on the concepts of ensuring that citizens should always utilize the facilities provided to them and integrating the sanitation programs with the rest of the essential services such as clean water and quality health care. This is where the importance of public awareness campaigns, programs that tend towards inculcation of a change in behaviour and constant monitoring comes into the limelight. It reaps a lot of benefits on investment made in these fields of complements- not only in the immediate but in the long-term too since it sets the groundwork to the long-term positive outcomes. The state of Tamil Nadu is required to have less of a generalized approach and more relational approach towards the gaps that exist in the current disparities in health and to make the most out of its efforts to change the situation in sanitation. This requires more than the infrastructure development strategies. The state should target education, engage the community, and promote education to ensure more development. The policies and plans highlighted in this paper can give the policy makers the road map which they can rely on in ensuring that all the

investments made in the field of public health are utilized optimally such that the resources are spread efficiently and equitably across all the districts.

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